

Detection and Prevention of Spoofing Attacks in the Internet of Underwater Things: A Comparative Review of Learning-Based Physical-Layer Authentication Techniques

Sehrish Siddique and Tatheer Zahra

Abstract—The Internet of Underwater Things (IoUT) is an underwater sensor network for sensing, communication and environmental monitoring in underwater environments, consisting of underwater sensors, autonomous underwater vehicles (AUVs), surface sinks, and monitoring stations. Conventional terrestrial IoT deployments are not subject to the same challenges as IoUT, particularly underwater acoustic communication, where bandwidth is limited, propagation delay is long, multipath interference is a significant issue, BER is high, nodes are mobile, and energy budgets are constrained. All these properties render it much more difficult to enforce strong security in underwater networks. The spoofing and impersonation attacks are particularly harmful among the threats exploiting such vulnerabilities since a malicious entity can make itself appear as a legitimate node, introduce false sensor data inside the network, or spoof sensor data or change routing and monitoring decisions in the network. In this paper, a representative set of ten papers is systematically reviewed in detail with a focus on physical-layer authentication (PLA) to provide a clear comparative overview of how these techniques are applied to protect underwater acoustic networks in the context of Internet of underwater things (IoUT). The work does not propose a new algorithm, but it takes a look at how current algorithms are able to extract and use channel-dependent features (e.g., channel impulse response, power delay profile, arrival delay, location consistency, time-reversal response, statistical channel parameters, and historical channel fingerprints) to confirm node identity at underwater nodes. The surveyed authentication techniques are quite wide ranging in terms of methodology, including reinforcement learning, support vector machines, recurrent neural networks, convolutional neural networks, cooperative authentication, time-reversal authentication, database-based authentication, differential-feature authentication and mobility-aware location tracking. The comparative analysis brings out meaningful trade-offs between these approaches for their adaptability, energy consumption, computational complexity, support for mobility, dependence on adversarial training data, and feasibility for real deployment. It also identifies a series of ongoing open challenges for the field, such as the lack of standardized benchmark datasets, lack of validation in real underwater scenarios, node mobility-induced channel variations, computational complexity, absence of adversarial data, scalability issues, and the general need for authentication systems that are lightweight, understandable, and compatible with cryptographic mechanisms.

Index Terms—Internet of Underwater Things, IoUT security, spoofing attack, impersonation attack, physical-layer authentication, underwater acoustic networks, machine learning, deep learning.

I. INTRODUCTION

The Internet of Underwater Things (IoUT) is essentially the expansion of networks to include the underwater environment. IoUT allows underwater sensors, autonomous underwater vehicles (AUVs), surface sink nodes, and remote monitoring stations to communicate with one another. Some examples of how IoUT is being utilized are marine observation, monitoring of off-shore structures, disaster warnings, surveillance, and collecting underwater data. Underwater nodes collect physical or environmental data and transmit this data to surface sink nodes. The surface sink nodes then relay the collected data to either land-based or cloud-based systems.

Due to differences in how acoustic waves propagate through water versus how radio frequency waves propagate through air, it will be impossible to extend terrestrial IoUT security to underwater IoUT security. There are many characteristics of underwater acoustic communications that limit its effectiveness. These characteristics include limited bandwidth, high latency, multipath fading, Doppler effects, high error rates in transmitting bits, node mobility and limited battery life. These characteristics also present unique challenges to implementing traditional security mechanisms in underwater environments. In order to effectively develop authentication methods for IoUT, it is necessary to consider the limitations associated with both security and underwater communication.

Spoofing and impersonation attacks threaten the identity and reliability of IoUT networks. During these attacks, an attacker attempts to imitate a legitimate underwater node by sending fraudulent identity information or by sending false sensing data/control messages. When the receiving node accepts the fraudulent message, the attacker can cause damage to the monitoring results, manipulate routing decisions, consume network resources and interfere with underwater activities. Spoofing attacks within IoUT environments pose a greater challenge than those in typical IoT environments because of the unique characteristics of underwater nodes (i.e. slow retransmissions, constant monitoring and excessive cryptographic exchanges).

A new promising method for detecting spoofing/impersonation attacks in underwater environments is Physical-Layer Authentication (PLA). Even though an attacker may successfully copy a legitimate node's

identification number or message format, they will find it extremely difficult to reproduce the exact acoustic channel properties between the legitimate sender and receiver. PLA authenticates a sender based on physical layer attributes of the channel. Examples of physical layer attributes that can be used for PLA include channel impulse response (CIR), power delay profile (PDP), arrival delay, statistical channel characteristics, spatial channel behavior, historical channel fingerprints, or time-reversal responses. Since these attributes depend largely upon the relative positions of the sender and receiver, the propagation paths taken by the wave, node mobility and environmental factors, they can be utilized as reliable authentication evidence for a particular link.

Although channels in underwater environments exist, they do not remain stable over time. As a result of node mobility, water motion, changes in environmental conditions, and time-varying multipath propagation, valid channels may exhibit varying fingerprints. A fixed authentication threshold is unlikely to work successfully in real world scenarios, especially when nodes are moving and there are natural fluctuations in the channel (such as underwater scenarios). The latter fact has encouraged the development of learning-based and adaptive physical-layer authentication (PLA) approaches. Some of the most significant contributions so far are on the side of SVM-based authentication without adversary training data [1], database-based PLA for dynamic underwater acoustic networks [2], differential CIR feature-based authentication [3], reinforcement learning and deep reinforcement learning for authentication parameter selection [4], RNN-based prediction for mobile underwater devices [5], CNN and location-tracking-based authentication [6], cooperative authentication using trusted nodes [7], time-reversal-based authentication [8], impersonation detection in line-of-sight underwater acoustic sensor networks [9], and mobility-aware PLA for underwater acoustic networks with mobile devices [10].

PLA detects when the physical-layer evidence that it receives from a sender does not match its expectations. Prevention happens at the receiver, sink or network controller level, when a decision to drop packets or nodes is taken after an unsuccessful authentication check. It is important to point out that it is not about preventing the attacker from transmitting, but preventing the legitimate network from receiving it. This separation is often ignored since most of the literature investigated tends to concentrate mainly on authentication or anomaly detection, without specifically addressing the prevention stage.

In this paper, a focused comparison of 10 papers published in the period from 2018 to 2025 is presented for the application of PLA techniques to detect the spoofing and impersonation attacks in underwater acoustic networks in the context of the wider Internet of Underwater Things (IoUT). Based on the following research questions, the review was organized:

- RQ1: What are the most frequently used physical-layer features for recognizing spoofing and impersonation in underwater IoUT networks?

- RQ2: What are the differences between these approaches, in terms of adaptability, energy consumption and computational complexity, and how do their learning-based mechanisms cope with node mobility?
- RQ3: Are the reviewed methods adversarial training data dependent or historical CIR dependent or do they depend on nodes for collaboration or do they depend on localization models or do they require real-world validation?
- RQ4: What are some open research challenges that must be overcome before spoofing defense mechanisms can be practically applied in real-world IoUT systems?

This paper has three main contributions. First, it categorizes the selected works according to the physical-layer characteristics that they take advantage of, as a framework for comparing spoofing and impersonation defenses. Second, it classifies the papers by their learning mechanisms and the nature of the authentication decisions they enable, and compares them with respect to network models, types of attacks, how mobile they are, what type of adversarial data they require, what metrics they use for evaluation, and their overall strengths and weaknesses. Thirdly, it highlights areas for additional research and future directions for practical IoUT spoofing defenses including lightweight learning, adaptive thresholds, benchmark data sets, mobility-aware authentication, hybrid PLA and cryptographic mechanisms, transfer learning, federated learning, and interpretable authentication decisions.

II. BACKGROUND OF IOUT NETWORKS

The Internet of Underwater Things (IoUT) refers to the systematic extension of networked sensing, communication and monitoring into the underwater space. A working IoUT architecture typically should integrate underwater sensor nodes, AUVs or any other mobile underwater devices, surface sinks and remote base stations into a coordinated operation. The configuration in this case is a collection of underwater sensors that reports physical and environmental data and the AUVs perform inspection, data collection, or relay tasks as needed. The data is typically sent via underwater acoustics links to a surface sink that then relays it to onshore systems, satellite, radio communications or cloud-based monitoring systems.

In this paper, IoUT is used as the umbrella application-layer concept that encompasses more specific network designations. The various instances of underwater acoustic network (UAN), underwater wireless acoustic network (UWAN), underwater acoustic sensor network (UASN), and underwater sensor network (UWSN) are related to the specific network models used in the studies reviewed. While these terms are similar, they are not synonymous. UASN and UWSN are more focused on sensor-node communications, while UWAN and UAN look at a broader scope of underwater communication configurations, including mobile acoustic devices and AUVs. The one major difference between underwater communication and terrestrial wireless communication is that radio frequency signals can not be transmitted very far underwater, forcing acoustic transmission to be the preferred method in most underwater applications. The dominance, however, has come

at a price. The use of acoustic links places strict and stringent requirements on any network around them — its bandwidth, propagation delay, multipath propagation, Doppler effects, environmental noise, high bit-error rate, and great sensitivity to the position of nodes, their operating depth, and the surrounding water conditions. Such challenges are exacerbated if nodes are mobile, as in AUV-supported or drifting-node scenarios [5], [10]. Energy limitation is also a major concern because underwater sensor batteries are difficult to replace or recharge after deployment [1].

The ease and difficulty of communication in the underwater acoustic channel is what makes it particularly relevant for security applications: the difficult aspects also provide evidence that can be used for the benefit of security applications. Physical-layer authentication (PLA) is based on a simple yet profound fact: the acoustical channel between a legal transmitter and receiver has channel-dependent properties that is not easily mimicked by a spatially separated adversary. Each of its features, such as channel impulse response (CIR), power delay profile (PDP), arrival delay, statistical channel behaviour, location consistency, time-reversal response and historical channel fingerprints contain identifiable information that can be used to determine where a transmission is coming from and can thus be used as a means of verifying who actually transmits. For instance, PDP is used in learning-based PHY-layer authentication [4], CIR-related features are used in database-based and differential-feature PLA [1], [2], [3], location behavior is used for mobile authentication [6], and time-reversal response is used as a multipath-based link signature [8].

The overall contribution of the underwater acoustic channel to the security of IoUT is defined by this duality. The very same channel variability that makes conventional security systems so frustrating — by limiting the bandwidth, allowing the delay, and munching the energy that frequent retransmission and copious amounts of authentication exchanges would consume — is what gives PLA the discriminating evidence it requires. The same applies to channels that are hard to work with: they're hard to impersonate. This balance between obstacle and resource makes PLA particularly suited for underwater acoustic networks in the context of IoU, and is why it is being considered as a defence mechanism against spoofing.

III. SPOOFING ATTACKS IN IOUT NETWORKS

In IoUT, a Spoofing attack is one that the malicious transmitter tries to present itself as a legitimate underwater node. This can manifest in many ways, such as spoofing the node id, sending messages with trusted id, sending fake sensor data or sending fake control messages. The related term impersonation is often used when the attacker deliberately impersonates an authorised transmitter. In this review, both spoofing and impersonation are considered as identity-deception attacks: attacks in which the identity asserted in an incoming packet is not consistent with the identity information observed by the receiver at the physical layer.

The problem of spoofing in IoUT is directly proportional to the level of criticality of its applications. Whether for environmental monitoring, surveillance, early-warning systems, or critical infrastructure, underwater networks are used regularly and where inaccurate or altered information can have profound impacts. A receiving entity or sink that may be fooled into believing that it received a valid packet could behave based on inaccurate environmental measurements, follow misdirected routing decisions or control decisions, use bandwidth and energy resources in the processing of bogus packets, and grow increasingly distrustful of the network's outputs. In addition to this, the environment in which underwater nodes operate is often remote and hard to reach, with low bandwidth and energy budgets, and long delays. In such an environment, the constant retransmission, often handshaking for keys and heavy encryption that can be done on terrestrial networks would become impractical.

A spoofing attack on an identity is based on the logical identity versus physical origin relationship. A malicious node can be able to duplicate an identity of a legitimate node and/or it can be able to duplicate an underwater acoustic channel between a legitimate node and the receiver, but reproducing a legitimate channel between the legitimate node and the receiver is a much harder problem. This is the exact space where PLA's purpose lies. A receiver with an identified name that receives a packet with the same name and the same CIR, PDP, delay behaviour, time-reversal response, location estimate, and historical channel pattern is not to be blamed if he or she suspects that the packet is not genuine. Aman et al. studied impersonation detection in line-of-sight underwater acoustic sensor networks [9], while later works examined spoofing and authentication under learning-based, cooperative, mobile, database-based, and time-reversal settings [1], [4], [5], [7], [8].

The attacker model varies across the reviewed papers. Some studies consider impersonation in line-of-sight underwater acoustic sensor networks [9]; some focus on spoofing detection through learned authentication parameters [4]; some consider resource-constrained networks without adversary training data [1]; and others address mobile underwater devices whose channel features naturally evolve over time [5], [6], [10]. Co-operative authentication further assumes that multiple trusted nodes can assist the sink in verifying a suspicious message [7]. Combined, these differences indicate that spoofing defence in IoUT is not a well-defined problem with a single solution; rather, it is influenced by the network model used, the attacker's location, node mobility, the existence of channel history, and the authentication architecture(s) that are being used.

One last word about terms is in order before proceeding. In this paper, detection refers to the process of recognizing that a mismatch between the identity asserted by the received transmission, and the evidence collected at the physical-layer received signal processing. The use of the word prevention is used in its strict sense: once a bad decision is made by PLA, the receiver, sink, or network controller can refuse to receive

the offending packet, ignore its contents, isolate the suspected node, or ask for re-authentication. This is not to say that you can stop the opponent from sending – software-layer means cannot stop the opponent from sending. It makes for ensuring that the spoofed transmission is not trusted or acted upon in the IoUT system.

Table I key elements in the threat model for spoofing that were explored in this review are summarised in Table I and include claimed identity, physical origin, capability of the attacker, potential impact, and the evidence used for authentication that comes from the PLA.

IV. PHYSICAL-LAYER AUTHENTICATION FOR SPOOFING DEFENSE

Physical-layer authentication (PLA): Authentication of a transmitter based on physical properties of the communication link that can be obtained directly from the link, instead of relying on higher-layer identifiers, passwords, or cryptographic credentials. This tends to be especially relevant to underwater acoustic networks used in IoUT, as copying the identifier of a legitimate node may be relatively easy, but reproducing the exact acoustic channel characteristics between the legitimate transmitter and its receiver is a much more complex problem. Those properties are dependent on transmitter-receiver geometry, propagation path, multipath structure, mobility of nodes, water conditions and the past history of the link's performance, which cannot be easily repeated by a spatially displaced adversary.

A normal PLA process consists of five steps, which are signal receiving, physical-feature extraction, decision processing, authentication decision and security action. This pipeline is summarised in Table II. What it does instead is provide the receiver, sink, or network controller with the ability to reject, mark, isolate, or re-authenticate any packet and/or node that has physical-layer evidence that doesn't match its claimed identity.

The physical features used in PLA change from study to study, as seen in the studies reviewed here. Channel impulse response (CIR) represents the multipath nature of the acoustic link and can be used as a link fingerprint: If the CIR derived feature does not correlate with the CIR derived feature of a legitimate node over a period of time, the packet might be marked as suspicious. Power delay profile (PDP) is a representation of the distribution of received signal power over multipath delays, and has been used in learning-based PHY-layer authentication for underwater sensor networks [4]. Arrival-delay and location-consistency features are useful when the receiver can relate the claimed identity to an expected position, trajectory, or propagation delay [6]. Statistical channel parameters can also be used in cooperative authentication, where trusted nodes locally evaluate channel-based evidence and send authentication beliefs to a sink for fusion [7].

Time-reversal PLA does just the opposite, and uses the multipath nature of underwater acoustic channels as an authentication resource. Instead of the multipath as a commu-

nication problem, time-reversal processing takes advantage of the fact that a legitimate link generates a focusing or resonating response. In underwater acoustic sensor networks, Zhao et al. use maximum time-reversal resonating strength (MTRRS), in combination with correlation in CIR database as the link-dependent evidence for node authentication in underwater acoustic sensor networks [8]. It is related, but not the same: Time-reversal PLA and CIR-based PLA: Time-reversal PLA uses the multipath focusing behaviour of the channel as the authentication signature while CIR-based PLA compares channel fingerprints directly.

A drawback of static fingerprinting is that underwater acoustic channels vary over time, which is something that cannot be handled using historical and differential channel features. Database-based PLA stores historical CIR or link-fingerprint information and compares current observations with updated channel records [2]. Differential feature-based PLA further compares current CIR samples with historical CIRs from the same link and extracts differential features that capture fine-grained link variation under time-varying underwater conditions [3]. Both techniques are not limited to the use of fixed fingerprints, but impose their own requirements: The management of the database, ageing of channels, storage space requirements, and the reliability of the observations in history.

To attend to specific techniques, it is important to consider a conceptual difference first. Learning based PLA does not necessarily mean learning to spoof behaviour. It learns in many cases the characteristics of a legitimate channel behaviour, an authentication threshold, a classifier boundary, a mobility prediction model or a feature-fusion rule. This distinction is essential because several underwater authentication methods are designed for situations where real adversary training data is unavailable or impractical to collect [1].

V. LEARNING-BASED DETECTION AND PREVENTION TECHNIQUES

A. *Threshold-Adaptive RL-Based PLA*

When uncertainties exist about the channel and the behaviour of the attacks by the receiver, and the receiver has to decide on the authentication parameters, then the use of reinforcement learning (RL)-based PLA becomes relevant. Xiao et al. use channel evidence in the form of PDP and optimise the authentication threshold using RL as well as deep reinforcement learning (DRL), [4]. This threshold represents a very basic trade-off: If the threshold is too low, legitimate packets may be rejected when the underwater channel fluctuates naturally; if the threshold is too high, then packets that are spoofed may be accepted even though they aren't legitimate. The RL agent learns to deal with this trade-off by making repeated interactions with the environment, and modifying its authentication policy based on feedback associated with authentication utility, detection behaviour, and previous decisions. DRL takes this approach to a larger state space and one in which the sink or the surface side controller has enough capacity to support deeper learning models.

TABLE I
THREAT MODEL FOR SPOOFING AND IMPERSONATION IN IOUT NETWORKS

Attack Aspect	Meaning in IoUT	PLA Evidence/Check
Claimed identity	The attacker uses or imitates the identifier of a legitimate underwater node.	Compare the claimed identity with channel-dependent features associated with the legitimate link.
Physical origin	The attacker transmits from a different position, depth, or propagation path than the legitimate node.	Check CIR, PDP, delay behavior, location consistency, or time-reversal response.
Target	The attack may target a receiver, surface sink, trusted helper node, or monitoring process.	Verify the received packet before accepting it into routing, sensing, or monitoring decisions.
Attacker capability	The attacker may forge packet identity or format, but may not be able to reproduce the exact acoustic channel fingerprint of the legitimate transmitter.	Detect mismatch between logical identity and physical-layer evidence.
Potential impact	Spoofed packets may inject false readings, mislead routing, waste bandwidth and energy, or reduce trust in the network.	Flag suspicious packets and prevent them from being accepted as trusted data.
Detection evidence	The receiver observes physical-layer features that are linked to the propagation environment and transmitter-receiver geometry.	Use CIR, PDP, arrival delay, statistical channel parameters, location estimates, time-reversal response, or historical fingerprints.
Prevention action	The system does not physically stop the attacker from transmitting; it rejects or isolates suspicious packets or nodes after failed authentication.	Drop, ignore, flag, isolate, or request re-authentication after a failed PLA decision.

TABLE II
GENERAL PLA PIPELINE FOR IOUT SPOOFING DEFENSE

Stage	Function and Example Evidence	Security Role
Signal reception	The receiver obtains a packet or acoustic observation from a claimed underwater transmitter.	Provides raw authentication evidence.
Feature extraction	Channel-dependent features such as CIR, PDP, arrival delay, channel statistics, location estimate, time-reversal response, or historical CIR pattern are extracted.	Converts the acoustic link into a physical fingerprint.
Decision processing	The system compares, classifies, predicts, or fuses the extracted evidence using methods such as SVM, RL/DRL, RNN/LSTM, CNN localization, database correlation, or cooperative belief fusion.	Checks whether evidence matches the claimed identity.
Authentication decision	The packet or node is classified as legitimate, suspicious, or unauthenticated using a threshold, prediction-error test, database match, GLRT, or belief-fusion decision.	Detects identity-channel inconsistency.
Security action	The network accepts, rejects, flags, isolates, ignores the packet, or requests re-authentication.	Provides authentication-based rejection or prevention.

The major advantage of this method is that it is adaptive, and as such, adapts to dynamic underwater acoustic environments that cannot be done with fixed-threshold methods. The downside, however, is evident: training requirements, how the parameters are tuned, the convergence behaviour and computational cost are all factors that need to be carefully evaluated before deployment in a resource-constrained IoUT system.

B. Classifier-Based PLA Without Adversary Data

SVM-based PLA is a rephrasing of the authentication problem as a classification problem. As it is common in most underwater security tasks, one of the main challenges with Zhao et al. is the absence of real adversary training data in underwater acoustic networks, as gathering attack samples is expensive, dangerous, and is often very task and

environment dependent [1]. Their approach involves database-correlative features based on CIR and generating a useful training set without the need for attacker samples. The SVM then learns a decision boundary that can separate the legitimate transmissions from the suspicious ones.

What makes this approach significant for IoUT is the fact that there is a lack of data that is adversarial to it. It is also more convenient to fit into resource-limited networks than techniques requiring extensive attack data or continuous online learning. Despite this, its validity will be dependent on the quality of the CIR databases it is dependent upon, and whether the patterns of links stored remain representative as the underwater channel evolves.

C. Sequence-Prediction-Based PLA for Mobility

The mobile underwater devices pose a unique authentication problem: legitimate channel features change over time as the transmitter moves, and a static fingerprint will result in false alarms with nodes that are merely carrying out their duty. To directly tackle this, Ardizzon et al. propose the application of an RNN/LSTM-based prediction to underwater acoustic networks with mobile devices [5]. The model is used to forecast the development of the channel related features over time and then authentication is effected by whether the channel behaviour observed in the feature occurs within the limits of the channel behaviour expected from the model.

Here, the authentication signal isn't just a feature value, but the way it changes over time. As the prediction error increases it becomes abnormal and the receiver can assume the packet or transmitter is suspicious. Since a generalised likelihood ratio test (GLRT) for the authentication decision is used in the source paper, that detail is included here for this reason. The key advantage of the method is that it is mobile aware, while the key disadvantage is that RNN/LSTM models have their own requirements: appropriate training data, careful tuning and non-trivial computational resources.

D. Location-Estimation-Based PLA

A location based PLA verifies a transmitter by asking whether the location or movement of a transmitter matches its claimed identity. Ventura et al. [6] suggest a location tracking authentication scheme with a CNN-based localisation model for estimating the location of the underwater source and a Kalman filter or RNN for projecting the source's location forward. The degree of mismatch between estimated and predicted location behaviour is then used to decide the authentication.

The conceptual solution seems to deal with mobile IoUT in a relatively sophisticated manner—that is, mobility is a condition rather than a red flag. However, its accuracy is linked to the quality of localisation, the accuracy of the environmental and/or propagation model, and reflectivity of the training/simulation environment to real deployment conditions. Since this work is an arXiv preprint at this time in the reference list, the claims should be treated with care and verified with any peer-reviewed version that becomes available.

E. Cooperative Multi-Receiver PLA

On the other hand, cooperative authentication is a method of enhancing PLA by leveraging the fact that the verification is spread over multiple trusted nodes, instead of being centralized into a single receiver. Diamant et al. propose a cooperative authentication method in which trusted nodes evaluate local beliefs based on underwater acoustic channel parameters, and the sink fuses these beliefs to make a final authentication decision [7]. The logic is that an attacker might be able to fool one receiver but it should be more difficult to maintain physically-layer evidence over multiple trusted receivers located at different points.

Cooperative PLA offers benefits of better reliability from spatial diversity and multi-node verification. The cost is the coordination overhead, meaning that there needs to be some nodes in the network that they trust and know how to reach, as well as know what to say, in order to get their beliefs to the sink in time. Cooperative PLA is thus best suited when network resources and trusted infrastructure are available to support it.

F. Time-Reversal and Database-Correlation PLA

Time-reversal PLA enhances underwater acoustic channels based on the multipath structure of the channel, rather than treating it as a communication burden, as a link signature. Zhao et al. use a technique called maximum time-reversal resonating strength (MTRRS) to characterize time-reversal resonating behavior and utilize time-reversal resonating strength correlation with a CIR database for node authentication [8]. The acoustic channel in this context is considered a security asset: the legitimate link has a characteristic multipath response that the attacker creates a different response in a spatially separated channel.

The method is firmly based on the physical facts of multipath propagation in the underwater environment, a very important asset. However, it can only be reliable when the

CIR is estimated accurately, the database is built correctly and the channel is tracked consistently. If the information stored becomes stale or channel estimation is poor, the authentication performance will be as well. Time-reversal PLA has potential, but requires rigorous testing in the dynamic conditions that can occur during actual underwater deployments.

G. Database-Based and Differential Historical-CIR PLA

The two PLA approaches that rely on historical channel information are not the same, and should not be confused, as it is called database-based PLA and differential feature-based PLA. Shi et al. propose database-based PLA for dynamic underwater acoustic networks by constructing and updating CIR-based databases to track time-varying channel behavior [2]. The goal is to compare current link observations with stored historical patterns so that authentication remains reliable even when the underwater channel changes.

Zhao et al. later develop differential feature-based PLA, where current CIR samples are compared with historical CIRs from the same link, and differential channel features are extracted to capture fine-grained changes in the acoustic channel [3]. This approach is more specific than general database matching because it focuses on differential relationships between current and historical channel observations. Its strength is robustness to time-varying underwater conditions, while its limitations include feature-management complexity, dependence on historical CIR quality, and the need to keep authentication evidence updated.

Overall, learning-based and feature-driven PLA methods contribute to spoofing detection by identifying inconsistency between claimed identity and physical-layer evidence. Their prevention role is operational: suspicious packets or nodes may be rejected, ignored, isolated, or sent for re-authentication after a failed PLA decision. This authentication-based rejection is useful for IoUT defense, but it should not be overstated as physically preventing the adversary from transmitting.

VI. SYNTHESIS OF REVIEWED LITERATURE

This section synthesizes the ten reviewed studies according to their authentication problem, physical-layer evidence, decision mechanism, evaluation basis, and deployment assumptions. Instead of treating the papers as isolated summaries, the review groups them into six technical directions: statistical impersonation detection, adaptive learning-based authentication, adversary-data-free classification, mobility-aware authentication, cooperative authentication, and CIR/time-reversal/database-based authentication.

A. Early Statistical and Hypothesis-Testing Authentication

Aman et al. address impersonation detection in line-of-sight underwater acoustic sensor networks [9]. Their work considers an active adversary that attempts to inject malicious data by impersonating a legitimate sensor node. The sink exploits physical observations such as distance, angle of arrival, and location-related estimates as device fingerprints. The authentication problem is formulated through binary hypothesis testing

for impersonation detection and maximum-likelihood-based tests for transmitter identification. This work is important because it directly models underwater impersonation rather than simply adapting a terrestrial IoT authentication model. However, its applicability is constrained by its line-of-sight assumption, analytical modeling assumptions, and simulation-based evaluation. Therefore, it provides an important baseline for underwater spoofing detection but does not fully address mobility, non-line-of-sight multipath, or highly dynamic IoUT deployments.

B. Learning-Based Adaptive Authentication

Xiao et al. propose a learning-based PHY-layer authentication framework for detecting spoofing attacks in underwater sensor networks [4]. The work uses the power delay profile (PDP) of the underwater acoustic channel as physical-layer evidence and applies reinforcement learning (RL) and deep reinforcement learning (DRL) to optimize authentication parameters. The central contribution is adaptive threshold or parameter selection under uncertain channel and attack conditions. This is significant because static thresholds may fail when underwater acoustic channels vary due to multipath, noise, and mobility. The method also highlights the false-alarm versus missed-detection trade-off: stricter authentication may reject legitimate packets, whereas looser authentication may accept spoofed ones. Its main limitation is the computational and training cost associated with RL/DRL-based adaptation, especially for resource-constrained underwater nodes. Exact numerical performance values should be taken directly from the original paper if required; otherwise, the comparison should remain qualitative.

C. Resource-Constrained and Adversary-Data-Free PLA

Zhao et al. specifically address PLA in resource-constrained underwater acoustic networks without requiring adversary training data [1]. This is a practically important problem because collecting real underwater spoofing samples is difficult, expensive, and environment-dependent. The proposed approach uses database-correlative CIR features and constructs training samples without relying on real adversary data. An SVM classifier is then used to support distributed node authentication. This makes the work highly relevant to IoUT because it targets both adversary-data scarcity and resource limitations. Its strength is that it reduces dependence on attack samples, while its limitation is that it still depends on representative legitimate CIR records and reliable database construction. If the channel database becomes outdated or does not represent current underwater conditions, authentication performance may degrade.

D. Mobility-Aware Authentication

Mobility-aware PLA is necessary because legitimate underwater devices such as AUVs, drifting sensors, and mobile relays naturally experience changing channel fingerprints. Casari et al. study PLA with mobile underwater acoustic devices and use the power-weighted average delay of channel

taps as an authentication feature [10]. Their approach uses a Kalman-filter-based method to track feature evolution and detect suspicious deviations. This work shows that mobility should not automatically be treated as an attack; instead, authentication should track expected feature variation over time.

Ardizzon et al. extend this direction through an RNN/LSTM-based approach for physical-layer authentication with mobile underwater devices [5]. The method extracts channel features and predicts their temporal evolution. Authentication is then based on whether observed behavior is consistent with the predicted legitimate channel evolution, with a generalized likelihood ratio test (GLRT) used for the decision process in the source paper. The strength of this approach is that it directly models temporal channel dynamics, while its limitation is dependence on suitable training data, computational resources, and environmental representativeness.

Ventura et al. propose a location-tracking authentication approach for underwater acoustic networks [6]. Their method first estimates the position of the underwater device using a CNN-based localization model and then predicts future location using a Kalman filter or RNN. Authentication is based on the mismatch between predicted and estimated position. This approach is well aligned with mobile IoUT scenarios because it uses trajectory consistency as authentication evidence. However, the method depends on localization accuracy, environmental calibration, and simulation assumptions. Since the cited work is currently listed as an arXiv preprint, its results should be treated as source-specific and verified against a final peer-reviewed version if available.

E. Cooperative Authentication

Diamant et al. propose cooperative authentication in underwater acoustic sensor networks [7]. Instead of relying on a single receiver, the method uses trusted nodes that independently evaluate channel-based authentication beliefs. These beliefs are then fused at the sink to reach a final authentication decision. The key insight is spatial diversity: an attacker may imitate a legitimate channel toward one receiver, but it is much harder to produce consistent physical-layer evidence across several trusted receivers located at different positions. This makes the method strong for multi-receiver authentication settings. However, its deployment requires trusted helper nodes, coordination, and additional communication overhead. Therefore, cooperative authentication is more suitable when infrastructure and energy resources allow collaboration.

F. Time-Reversal, Database-Based, and Differential CIR Authentication

Zhao et al. propose a time-reversal-based PLA method for underwater acoustic sensor networks [8]. The method uses maximum time-reversal resonating strength (MTRRS) and CIR database correlation as authentication evidence. Its importance lies in converting underwater multipath propagation from a communication impairment into a security feature. If the

received channel behavior does not produce the expected time-reversal resonating response for the claimed link, the node may be treated as suspicious. The method is physically grounded and includes simulation and sea-trial CIR-based verification in the source paper, but it also depends on reliable CIR estimation, database maintenance, and channel tracking.

Shi et al. address database-based PLA for dynamic underwater acoustic networks [2]. Their work stores and updates historical CIR/link-fingerprint information so that the authentication system can compare current observations with previous legitimate channel behavior. This approach directly addresses the problem that underwater channel fingerprints are not perfectly static. Its main limitation is database aging: stored channel records may become less representative as the acoustic environment changes.

Zhao et al. later propose differential feature-based PLA for underwater acoustic sensor networks [3]. Instead of relying only on static channel signatures, the method compares current CIR samples with historical CIR samples from the same link and extracts multidimensional differential features. This supports authentication under time-varying underwater conditions. The method is stronger than simple static-fingerprint matching because it explicitly models link variation, but it requires reliable historical CIR samples, feature management, and careful evaluation under changing ocean environments.

Overall, the reviewed literature shows that IoUT spoofing defense is not solved by a single method. Statistical detection is analytically clear but often assumption-dependent; RL/DRL is adaptive but computationally heavier; SVM-based PLA reduces the need for adversary data but still needs reliable legitimate channel records; RNN/CNN and location-tracking methods are useful for mobility but depend on training and localization quality; cooperative authentication improves reliability through spatial diversity but adds overhead; and time-reversal, database-based, and differential CIR methods exploit rich underwater channel structure but require accurate and updated channel information.

VII. COMPARATIVE ANALYSIS

The reviewed methods differ significantly in terms of attack model, physical-layer evidence, learning mechanism, evaluation basis, mobility support, adversary-data requirement, and deployment assumptions. For this reason, the comparison is divided into three complementary views. Table III presents the core technical evidence matrix, including the main problem addressed, attack model, PLA feature, decision method, and main strength of each reviewed study. Table IV then compares deployment-oriented factors such as adversary-data requirement, mobility support, evaluation type, main limitation, and deployment assumption. Finally, Table V summarizes the major PLA method families at a higher level using qualitative ratings.

Table III shows that the reviewed studies do not address the same technical problem. Some works focus on impersonation detection under line-of-sight assumptions [9], while others address adaptive spoofing detection [4], adversary-data scarcity

[1], mobile-device authentication [10], [5], [6], cooperative verification [7], time-reversal authentication [8], and historical or differential CIR-based authentication for dynamic channels [2], [3]. This confirms that IoUT spoofing defense is not a single-method problem; rather, each approach is designed around a different physical-layer feature, network model, and decision mechanism.

Table IV highlights the practical differences among the reviewed studies. The strongest mobility-oriented methods are the mobile-device and location-tracking approaches [10], [5], [6], because they treat channel variation as an expected consequence of movement rather than automatically treating it as an attack. Adversary-data scarcity is addressed most directly by Zhao et al. [1], which is important because real underwater attack samples are difficult to collect. Cooperative authentication [7] and time-reversal PLA [8] rely strongly on physical deployment assumptions: trusted helper nodes in the former and reliable CIR/time-reversal response estimation in the latter. Database-based and differential CIR methods [2], [3] are more suitable for dynamic channel environments than purely static fingerprint methods, but they introduce database maintenance and historical-record quality challenges.

Table V provides a high-level qualitative comparison of the major PLA method families. This table should not be interpreted as a direct numerical ranking because the reviewed papers use different evaluation metrics, channel models, attack assumptions, datasets, and validation settings. For example, analytical hypothesis testing, Bellhop simulation, sea-trial CIR analysis, and sea experiments are not directly equivalent. Therefore, a method that performs well in one validation setting cannot automatically be assumed to outperform another method evaluated under a different setting.

From a deployment perspective, no reviewed method dominates all others. RL/DRL and RNN/CNN-based methods provide adaptability but increase computational demand. SVM-based authentication without adversary data is attractive for resource-constrained environments, but depends on representative legitimate CIR data. Cooperative authentication increases reliability but requires trusted nodes and communication overhead. Time-reversal and database/differential CIR methods are physically meaningful in multipath-rich underwater channels, but their performance depends on channel estimation, database freshness, and environmental stability. Therefore, practical IoUT spoofing defense will likely require hybrid designs that combine lightweight learning, adaptive thresholds, channel-history management, and authentication-based rejection while keeping energy and communication overhead within underwater network limits.

VIII. TAXONOMY OF REVIEWED TECHNIQUES

The reviewed studies cannot be classified accurately using a single flat taxonomy because several techniques overlap in practice. For example, database-based PLA, differential-feature PLA, and time-reversal PLA may all use CIR-related evidence, but they differ in how the evidence is processed and in what deployment condition they target. Therefore, this

TABLE III
CORE EVIDENCE MATRIX OF REVIEWED PLA-BASED SPOOFING DEFENSE STUDIES

Ref.	Study	Main Problem	Attack Model	PLA Feature	Decision Method	Main Strength
[9]	Aman et al.	Impersonation detection in LoS UASN	Active node impersonation	Distance, AoA, location estimates	Hypothesis testing and ML estimation	Direct underwater impersonation model
[4]	Xiao et al.	Adaptive spoofing detection	Spoofing in UWSN	PDP	RL/DRL parameter selection	Adaptive authentication threshold/utility optimization
[1]	Zhao et al.	PLA without adversary training data	Spoofing in resource-constrained UAN	Database-correlative features	CIR SVM classification	Addresses adversary-data scarcity
[10]	Casari et al.	PLA with mobile devices	Mobile impersonating transmitter	Power-weighted average delay	Kalman-filter-based tracking	Handles mobile channel evolution
[5]	Ardizzon et al.	RNN-based mobile PLA	Mobile impersonating transmitter	Temporal channel-feature evolution	LSTM prediction and GLRT	Models temporal mobility patterns
[6]	Ventura et al.	Location-tracking authentication	Source/location spoofing	UWAC/location behavior	CNN localization with Kalman/RNN prediction	Uses trajectory consistency
[7]	Diamant et al.	Cooperative authentication	Fake legitimate messengers	Statistical channel parameters	Trusted-node belief fusion	Uses spatial diversity
[8]	Zhao et al.	Time-reversal node authentication	Spoofing in UASN	MTRRS and CIR correlation	Time-reversal decision	Uses multipath as security evidence
[2]	Shi et al.	Database-based PLA for dynamic channels	Unauthorized/spoofed nodes	Historical CIR/link fingerprints	Database matching/correlation	Tracks dynamic channel fingerprints
[3]	Zhao et al.	Differential-feature PLA	Spoofing in dynamic UASN	Differential CIR features	Differential-feature classification	Robust to time-varying channel behavior

TABLE IV
DEPLOYMENT AND EVALUATION CHARACTERISTICS OF REVIEWED PLA STUDIES

Ref.	Needs Adversary Data?	Mobility Support	Evaluation Type	Main Limitation	Deployment Assumption
[9]	Not central	Low	Analytical modeling and simulation	LoS and model assumptions	Receiver can estimate physical/location-related parameters
[4]	Used in evaluation setting; exact dependency needs source check	Limited	Simulation-based evaluation; exact details need source verification	Training and computation cost	Sink/controller can support RL/DRL adaptation
[1]	No real adversary data required	Low–Medium	Simulation and sea-trial CIR analysis	Needs reliable legitimate CIR database	Legitimate CIR records are available and representative
[10]	Not central	High	Bellhop-generated simulation dataset	Depends on mobility and feature-tracking assumptions	Mobility model reasonably reflects device movement
[5]	Not central	High	Bellhop ray-tracing simulation data	Training and computational requirements	Training data represents mobile channel evolution
[6]	Not central	High	Bellhop-based simulation; arXiv preprint	Localization and model dependence	Localization model is accurate enough for authentication
[7]	Not central	Medium	Simulation and sea experiments	Requires trusted helpers and overhead	Trusted nodes are available for belief fusion
[8]	Not central	Low–Medium	Simulation and sea-trial CIR data	Needs CIR estimation and database maintenance	Time-reversal response can be reliably estimated
[2]	Not central	Medium	Simulation and sea-trial study	Database aging and storage overhead	Historical CIR database remains updated
[3]	Constructed labeled samples; exact adversary-data setup needs source check	Medium	Simulation and sea-trial CIR datasets	Needs reliable historical CIRs and feature management	Differential CIR features remain representative

review adopts a multidimensional taxonomy that separates the reviewed techniques into three analytical dimensions: physical-layer feature source, learning or decision mechanism, and deployment condition.

Fig. 1 summarizes this multidimensional view. The purpose of the taxonomy is not to claim that each paper belongs to only one category. Rather, it separates different aspects of a PLA method so that overlapping approaches can be compared more clearly. For instance, Zhao et al.’s time-reversal scheme uses MTRRS and CIR database correlation [8], while Shi et al.’s database-based PLA and Zhao et al.’s differential-feature PLA both rely on historical CIR behavior but differ in their authentication logic [2], [3]. Similarly, mobility-aware methods may use Kalman filtering, RNN/LSTM prediction, or CNN-based localization depending on whether the target is

channel-feature tracking or location consistency [5], [6], [10].

Table VI further explains the taxonomy by linking each dimension to representative techniques and its security relevance. This structure avoids treating feature sources, algorithms, and deployment assumptions as identical categories.

IX. CHALLENGES AND LIMITATIONS

Although the reviewed studies show that PLA is a promising direction for spoofing and impersonation defense in IoT-related underwater acoustic networks, several limitations remain before these methods can be considered deployment-ready. These limitations are not only algorithmic; they also involve datasets, evaluation fairness, mobility, energy cost, channel drift, attacker modeling, and deployment assumptions.

First, benchmark datasets for underwater acoustic security remain limited. Several reviewed works use analytical

TABLE V
QUALITATIVE COMPARISON OF PLA-BASED SPOOFING DEFENSE METHOD FAMILIES

Method Family	Accuracy Evidence	Adaptability	Energy Cost	Complexity	Mobility Support	Deployment Readiness
Hypothesis/ML detection	Qualitative comparison only; analytical and simulation evidence	Low–Medium	Low–Medium	Medium	Low	Useful baseline under LoS assumptions
RL/DRL-based PLA	Qualitative comparison only; source reports authentication utility/detection metrics	High	Medium–High	High	Medium	Adaptive but computationally demanding
SVM without adversary data	Qualitative comparison only; simulation and sea-trial CIR analysis	Medium	Low–Medium	Medium	Low–Medium	Practical for adversary-data scarcity if CIR database is reliable
RNN/LSTM mobile PLA	Qualitative comparison only; Bellhop simulation evidence	High	High	High	High	Strong for mobility but needs training and tuning
CNN/location tracking	Qualitative comparison only; Bellhop simulation in preprint	High	High	High	High	Promising but localization-dependent
Cooperative authentication	Qualitative comparison only; simulation and sea-experiment evidence	Medium	Medium–High	Medium–High	Medium	Useful where trusted helpers are available
Time-reversal PLA	Qualitative comparison only; simulation and sea-trial CIR evidence	Medium	Medium	Medium–High	Low–Medium	Physically grounded but CIR/TR estimation dependent
Database/differential CIR PLA	Qualitative comparison only; simulation and sea-trial CIR evidence	High	Medium	Medium–High	Medium	Suitable for dynamic channels but needs database maintenance

TABLE VI
MULTIDIMENSIONAL TAXONOMY OF REVIEWED PLA TECHNIQUES

Taxonomy Dimension	Category	Representative Techniques/Papers	Security Relevance
Physical-layer feature source	CIR / PDP	CIR-based SVM and database/differential features [1], [2], [3]; PDP-based RL/DRL authentication [4]	Uses channel response behavior as a link-dependent fingerprint for spoofing detection.
Physical-layer feature source	Location, AoA, and delay behavior	LoS impersonation detection [9]; location-tracking authentication [6]; mobile-device delay tracking [10]	Checks whether the claimed node identity is consistent with physical position, propagation behavior, or movement.
Physical-layer feature source	Time-reversal response	MTRRS-based node authentication with CIR database correlation [8]	Uses multipath focusing/resonating behavior as a physical signature of the legitimate link.
Physical-layer feature source	Historical and differential CIR	Database-based PLA [2]; differential-feature PLA [3]; adversary-data-free CIR features [1]	Supports authentication under time-varying channels by comparing current observations with historical link evidence.
Learning/decision mechanism	Hypothesis testing and ML estimation	Impersonation detection and transmitter identification [9]	Provides an analytically clear baseline for legitimate-versus-impersonator decisions.
Learning/decision mechanism	RL/DRL	Adaptive authentication parameter or threshold selection [4]	Adjusts the authentication policy under uncertain channel and attack conditions.
Learning/decision mechanism	SVM/classification	SVM-based PLA without real adversary training data [1]; differential-feature classification [3]	Learns a decision boundary from channel-derived features, including cases where adversary data is unavailable.
Learning/decision mechanism	RNN/LSTM, CNN, and belief fusion	RNN/LSTM mobile PLA [5]; CNN localization and Kalman/RNN prediction [6]; trusted-node belief fusion [7]	Supports mobility-aware prediction, location consistency, and cooperative authentication decisions.
Deployment condition	Resource-constrained nodes	Distributed PLA without adversary data [1]	Addresses limited bandwidth, energy, and training-data availability in underwater deployments.
Deployment condition	Mobile nodes	Mobile-device PLA [10]; RNN-based mobile PLA [5]; location tracking [6]	Reduces false alarms caused by legitimate movement of AUVs, drifting sensors, or mobile relays.
Deployment condition	Dynamic channels	Database-based PLA [2]; differential-feature PLA [3]; time-reversal PLA [8]	Handles channel variation by updating or comparing link fingerprints over time.
Deployment condition	Multi-receiver cooperative setting	Cooperative authentication using trusted nodes [7]	Uses spatial diversity and trusted helper nodes to make spoofing harder across multiple receivers.

modeling, simulation, Bellhop-generated channels, sea-trial CIR data, or sea experiments, but these evaluation settings are heterogeneous and not directly comparable. For example, hypothesis-testing methods may be evaluated analytically

or through simulation [9], mobility-aware methods may use Bellhop-based simulation data [5], [6], [10], and CIR-based or time-reversal methods may include sea-trial CIR evidence [1], [2], [3], [8]. Therefore, it is difficult to make a fair nu-

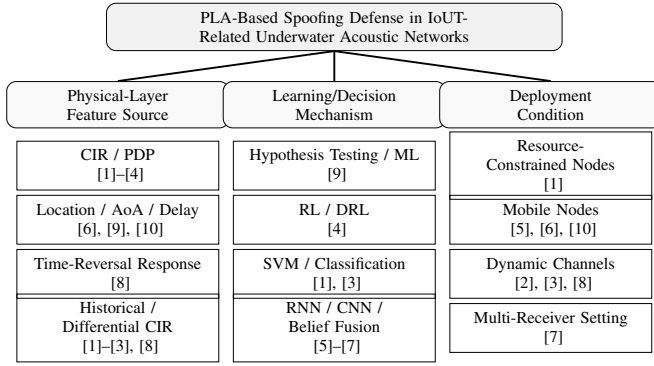


Fig. 1. Multidimensional taxonomy of reviewed PLA techniques for spoofing and impersonation defense in IoUT-related underwater acoustic networks.

merical comparison across methods without common datasets, common attack models, and shared evaluation metrics.

Second, real-environment validation remains uneven across the literature. It would be inaccurate to say that sea validation is entirely absent, because some works use sea experiments or sea-trial CIR data [1], [2], [3], [7], [8]. However, the validation conditions differ across papers, and results obtained under one channel model, location, mobility pattern, or sea-trial setting may not transfer directly to another underwater environment. This creates a reproducibility and generalization challenge.

Third, mobility creates a major authentication difficulty. Static PLA assumes that a legitimate transmitter-receiver link has a relatively stable fingerprint, but AUVs, drifting sensors, and mobile relays naturally change their channel features over time. Mobility-aware approaches using Kalman filtering, RNN/LSTM prediction, and location tracking address this issue [10], [5], [6]. However, prediction error, inaccurate localization, and environmental mismatch can still cause a legitimate moving node to appear suspicious, increasing the risk of false rejection.

Fourth, PLA must balance false acceptance and false rejection. A false acceptance occurs when a spoofed packet is treated as legitimate, while a false rejection occurs when a legitimate packet is rejected because of natural channel variation. This trade-off is especially important in underwater networks because retransmission and re-authentication may consume significant bandwidth and energy. RL/DRL-based authentication explicitly relates to threshold or parameter adaptation [4], while prediction-based and database-based approaches also depend on how much deviation is tolerated before a packet is flagged [2], [5].

Fifth, adversary-data scarcity remains a fundamental limitation. Realistic spoofing samples are difficult to collect in underwater environments because attacker positions, behavior, and channel conditions vary. Zhao et al. directly address this by proposing PLA without adversary training data [1]. However, broader evaluation is still needed to understand how adversary-data-free authentication behaves across different underwater environments, mobility patterns, and attacker strategies.

Sixth, energy and computational overhead remain deployment concerns. RL/DRL, RNN/LSTM prediction, CNN-based localization, cooperative belief fusion, and frequent database updates may improve authentication adaptability, but they also increase computational or communication cost [4], [5], [6], [7]. This is critical because underwater nodes often operate with limited batteries and low acoustic bandwidth. A method that is accurate in simulation may still be impractical if it requires excessive processing, communication, or memory.

Seventh, database aging and channel drift affect CIR-based, time-reversal, database-based, and differential-feature methods. Historical CIR records and link fingerprints may become outdated as water conditions, node positions, and multipath structures change. Database-based PLA and differential-feature PLA are designed to address dynamic channels [2], [3], but they still require reliable historical records and appropriate update mechanisms. Online updating may help, but it must avoid normalizing attacker behavior as legitimate channel drift.

Eighth, several methods rely on deployment assumptions that may not always hold in real IoUT systems. Cooperative authentication assumes the availability of trusted helper nodes and reliable communication with the sink [7]. Location-tracking authentication assumes sufficiently accurate localization and environmental calibration [6]. Time-reversal PLA assumes that MTRRS and CIR database correlation can be estimated reliably [8]. These assumptions should be made explicit when selecting a method for a practical deployment.

Finally, scalability remains an open concern. As IoUT networks grow to include more static sensors, mobile AUVs, surface sinks, and cooperative nodes, authentication must remain efficient in terms of energy, memory, processing delay, and communication overhead. Scalable PLA must support multiple links, changing topology, mobile devices, and database maintenance without overwhelming resource-constrained underwater nodes. Therefore, practical IoUT spoofing defense requires not only accurate detection but also lightweight, adaptive, and deployment-aware authentication design.

X. RECOMMENDATIONS

The reviewed literature suggests that practical IoUT spoofing defense should not rely on a single authentication technique. Instead, authentication design should be selected according to resource limits, mobility level, availability of historical channel data, and deployment assumptions. Table VII summarizes the main gaps identified in this review and maps them to recommended directions.

First, lightweight and energy-aware authentication should be prioritized for resource-constrained underwater nodes. RL/DRL, RNN/LSTM, CNN-based localization, and cooperative authentication may improve adaptability, but they can also increase computation, memory use, and communication overhead [4], [5], [6], [7]. Resource-aware designs such as adversary-data-free SVM-based PLA are therefore important for deployments where underwater nodes cannot support expensive processing or frequent communication [1].

TABLE VII
GAP-TO-RECOMMENDATION MAPPING FOR IOUT SPOOFING DEFENSE

Identified Gap	Recommended Direction	Relevant Methods/Papers	Deployment Rationale
Energy and computation constraints	Use lightweight and energy-aware authentication models.	Resource-constrained PLA [1]; RL/DRL and RNN/LSTM methods [4], [5]	Underwater nodes have limited battery and bandwidth, so authentication should minimize computation, memory, and communication overhead.
Time-varying underwater channels	Use adaptive thresholds and channel-history-aware authentication.	Database-based PLA [2]; differential CIR features [3]; RL/DRL threshold adaptation [4]	Static thresholds may produce false alarms when legitimate channels change because of mobility, multipath, or environmental variation.
Mobility of AUVs and drifting nodes	Design mobility-aware authentication using prediction and location consistency.	Mobile-device PLA [10]; RNN/LSTM prediction [5]; location tracking [6]	Legitimate movement changes channel fingerprints, so mobility should be modeled as an expected condition rather than always treated as suspicious.
Limited adversary training data	Prefer methods that do not require real spoofing samples when such samples are unavailable.	PLA without adversary training data [1]	Real underwater attack samples are difficult, costly, and risky to collect; relying on them may reduce practical deployability.
Heterogeneous evaluation settings	Improve benchmarking and report evaluation conditions clearly.	Analytical, Bellhop, simulation, sea-experiment, and sea-trial studies across [1]– [10]	Methods evaluated under different channel models and datasets cannot be fairly ranked without common metrics and comparable test conditions.
Single-layer security weakness	Combine PLA with lightweight cryptographic protection where feasible.	PLA methods across [1]– [10]	Cryptography can support integrity and confidentiality, while PLA verifies physical identity; the combination should remain lightweight.
Trusted-node and cooperation overhead	Use cooperative authentication only where helper nodes and communication overhead are acceptable.	Cooperative authentication [7]	Belief fusion can improve reliability, but trusted helpers, coordination, and extra communication may not be feasible in all IoUT deployments.
Low interpretability of learning decisions	Include explainable authentication outputs.	Learning-based and prediction-based PLA [4], [5], [6]	Operators should know whether rejection was caused by CIR mismatch, PDP deviation, location inconsistency, MTRRS abnormality, or prediction error.

Second, authentication thresholds and channel fingerprints should be adaptive rather than static. Underwater acoustic channels change because of multipath, mobility, water conditions, and node position. Database-based PLA and differential CIR approaches show that historical channel information can support authentication in dynamic environments [2], [3]. However, such methods should also include mechanisms for handling database aging and channel drift.

Third, mobility-aware authentication should be treated as a core requirement for AUVs, drifting sensors, and mobile relays. Mobile-device PLA, RNN/LSTM prediction, and location-tracking authentication show that movement can be modeled through feature evolution or trajectory consistency [10], [5], [6]. This reduces the risk of treating legitimate mobility as spoofing behavior.

Fourth, PLA should be considered as part of a layered security design rather than a complete replacement for cryptography. A layered design could combine cryptographic integrity and confidentiality with physical identity verification. This recommendation should be understood as a design direction, not as a solved result in the reviewed literature. The main challenge is to combine both layers without adding excessive packet overhead, delay, or computation.

Fifth, evaluation fairness should be improved through clearer benchmarking. The reviewed studies use different

evaluation settings, including analytical modeling, simulation, Bellhop-based channels, sea experiments, and sea-trial CIR data. These settings are not directly equivalent, so future comparisons should report evaluation conditions, attack assumptions, mobility models, and resource costs more explicitly.

Finally, learning-based PLA should include explainable decision support. If a node is rejected, the system should indicate whether the decision was caused by CIR mismatch, PDP deviation, abnormal time-reversal response, location inconsistency, or prediction error. Such interpretability would make PLA easier to debug, compare, and trust in practical IoUT deployments.

XI. FUTURE RESEARCH DIRECTIONS

Future research should first focus on benchmark datasets and reproducibility. Common datasets for underwater spoofing and PLA should include CIR, PDP, delay, location or trajectory information, mobility conditions, environmental context, and adversary-related samples where feasible. Without common benchmarks, it remains difficult to compare RL/DRL, SVM, RNN/LSTM, CNN, cooperative, time-reversal, database-based, and differential-feature methods under equivalent conditions.

Second, online and adaptive learning should be studied carefully. Underwater channels evolve over time, so authenti-

cation models may need to update after deployment. However, online adaptation must avoid normalizing attacker behavior as legitimate channel drift. This is especially important for methods that depend on historical CIR databases, adaptive thresholds, or mobility prediction [2], [3], [4], [5].

Third, cross-environment transfer learning should be investigated. A model trained in one underwater environment may not directly generalize to another environment with different depth, salinity, noise, temperature, geometry, or mobility patterns. Transfer learning could reduce retraining cost, but its reliability must be tested under realistic underwater channel variation.

Fourth, federated or distributed learning could be explored as a future direction for IoUT authentication. This is an inference-based future direction, not a proven result from the reviewed studies. In principle, distributed learning may allow multiple sinks or nodes to improve authentication models without sharing raw acoustic data centrally. However, communication overhead, synchronization, model poisoning, and underwater bandwidth limits must be carefully considered.

Fifth, future work should develop mobility-aware hybrid authentication. Such systems could combine channel fingerprints, location consistency, prediction models, and lightweight classification. For example, RNN/LSTM prediction can help track temporal channel evolution [5], while CNN or location-tracking methods can support trajectory consistency [6]. The goal should be to reduce false alarms for legitimate mobile nodes while still detecting impersonation.

Sixth, energy-aware model compression and lightweight inference should become a major design objective. Future learning-based PLA methods should report not only authentication performance but also computation time, memory requirement, communication overhead, and energy cost. These factors may determine whether a method is deployable in real IoUT systems.

Seventh, robustness against model drift, poisoning, and adaptive spoofers should be studied more directly. If attackers observe authentication behavior over time, they may try to adapt their transmissions or influence model updates. Future PLA systems should therefore include safeguards against learning corrupted patterns or accepting malicious behavior as normal.

Eighth, explainable PLA decision support should be developed. Authentication systems should provide interpretable reasons for rejection, such as CIR mismatch, PDP deviation, MTRRS abnormality, location inconsistency, or prediction error. This would support debugging, operator trust, and comparison between methods.

Finally, hybrid PLA and cryptographic security should be explored under underwater resource constraints. Cryptography can protect message integrity and confidentiality, while PLA can verify physical identity. The open research challenge is to combine both without creating excessive delay, packet overhead, or energy consumption in bandwidth-limited acoustic networks.

XII. CONCLUSION

This paper reviewed spoofing and impersonation defense in IoUT-related underwater acoustic networks, with a focus on learning-based and feature-driven physical-layer authentication. Spoofing is a serious security problem in IoUT because a malicious transmitter may forge a legitimate identity, inject false data, or mislead monitoring and routing decisions. At the same time, underwater acoustic networks face limited bandwidth, long propagation delay, multipath propagation, mobility, and energy constraints, making conventional authentication alone difficult to apply.

The reviewed studies indicate that PLA is a useful complementary defense because an attacker may copy an identifier or packet format, but cannot easily reproduce the exact acoustic channel behavior of a legitimate transmitter-receiver link. The reviewed methods use physical-layer evidence such as CIR, PDP, statistical channel parameters, arrival delay, location consistency, time-reversal response, historical channel fingerprints, and differential CIR features.

The ten selected studies represent several research directions: hypothesis-testing-based impersonation detection [9], RL/DRL-based adaptive authentication [4], SVM-based PLA without real adversary training data [1], mobility-aware PLA using Kalman filtering and RNN/LSTM prediction [10], [5], location-tracking authentication [6], cooperative authentication using trusted nodes [7], time-reversal PLA [8], database-based PLA [2], and differential-feature-based PLA [3]. These approaches support spoofing detection and authentication-based rejection, but they differ in assumptions, resource cost, mobility support, evaluation setting, and deployment readiness.

No single reviewed method dominates all deployment scenarios. Adaptive learning methods are flexible but may be computationally expensive. SVM-based PLA without adversary data is attractive for resource-constrained environments, but depends on representative legitimate CIR records. Mobility-aware methods address AUVs and drifting nodes, but require reliable prediction or localization. Cooperative authentication improves reliability through spatial diversity, but depends on trusted helper nodes and additional communication. Time-reversal, database-based, and differential CIR methods exploit underwater channel structure, but require reliable channel estimation and updated historical evidence.

The main contribution of this paper is therefore not a new algorithm, but a focused comparative review of representative PLA-based spoofing defense techniques. The paper organizes the literature through a taxonomy, compares method families, identifies deployment limitations, and outlines future research directions. Practical IoUT spoofing defense should move toward lightweight, mobility-aware, explainable, benchmarked, and hybrid authentication frameworks that combine physical-layer evidence with deployment-conscious security design.

REFERENCES

- [1] R. Zhao, T. Shi, C. Liu, X. Shen, and O. A. Dobre, "Physical layer authentication without adversary training data in resource-constrained underwater acoustic networks," *IEEE Sensors Journal*,

- vol. 23, no. 22, pp. 28 270–28 281, 2023. [Online]. Available: <https://doi.org/10.1109/JSEN.2023.3321777>
- [2] T. Shi, R. Zhao, X. Shen, and H. Wang, "Database-based physical layer authentication for dynamic underwater acoustic networks," in *OCEANS 2023 - Limerick*, 2023. [Online]. Available: <https://doi.org/10.1109/OCEANSLimerick52467.2023.10244704>
 - [3] R. Zhao, J. Li, T. Shi, and H. Wang, "Differential feature-based physical layer authentication for underwater acoustic sensor networks," *IEEE Sensors Journal*, 2025. [Online]. Available: <https://doi.org/10.1109/JSEN.2025.3613742>
 - [4] L. Xiao, G. Sheng, X. Wan, W. Su, and P. Cheng, "Learning-based phy-layer authentication for underwater sensor networks," *IEEE Communications Letters*, vol. 23, no. 1, pp. 60–63, 2019. [Online]. Available: <https://doi.org/10.1109/LCOMM.2018.2877317>
 - [5] F. Ardizzon, P. Casari, and S. Tomasin, "A rnn-based approach to physical layer authentication in underwater acoustic networks with mobile devices," *Computer Networks*, vol. 243, p. 110311, 2024. [Online]. Available: <https://doi.org/10.1016/j.comnet.2024.110311>
 - [6] G. Ventura, F. Ardizzon, and S. Tomasin, "Authentication by location tracking in underwater acoustic networks," arXiv preprint, 2024. [Online]. Available: <https://arxiv.org/abs/2410.03511>
 - [7] R. Diamant, P. Casari, and S. Tomasin, "Cooperative authentication in underwater acoustic sensor networks," *IEEE Transactions on Wireless Communications*, vol. 18, no. 2, pp. 954–968, 2019. [Online]. Available: <https://doi.org/10.1109/TWC.2018.2886896>
 - [8] R. Zhao, M. J. Khalid, O. A. Dobre, and X. Wang, "Physical layer node authentication in underwater acoustic sensor networks using time-reversal," *IEEE Sensors Journal*, 2022. [Online]. Available: <https://doi.org/10.1109/JSEN.2022.3142160>
 - [9] W. Aman, M. M. U. Rahman, J. Qadir, H. Pervaiz, and Q. Ni, "Impersonation detection in line-of-sight underwater acoustic sensor networks," *IEEE Access*, vol. 6, pp. 44 459–44 472, 2018. [Online]. Available: <https://doi.org/10.1109/ACCESS.2018.2863945>
 - [10] P. Casari, F. Ardizzon, and S. Tomasin, "Physical layer authentication in underwater acoustic networks with mobile devices," in *Proceedings of the 16th International Conference on Underwater Networks and Systems (WUWNet)*, 2022. [Online]. Available: <https://doi.org/10.1145/3567600.3567604>